

The updated assessment model for the revised sardine stock structure hypothesis

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This document details the stock assessment model of the revised South African sardine stock structure hypothesis. The model allows for two stocks (Warm Temperate Sardine and Cool Temperate Sardine), distributed between two areas (the west and south coasts of South Africa, separated at Cape Agulhas). Sardine growth is coast-dependent, and the distribution of length-at-age off the south coast is modified from the south coast growth curve to accommodate the immigration of individuals from the west coast which have grown according to a different growth curve for the first part of their life on the west coast. Catch by stock off the west coast can either be proportional to the numbers of exploitable sardine-at-age by stock off the west coast, or related to alternative pre-specified weightings of CTS to WTS between the region to the north of Cape Point and that between Cape Point and Cape Agulhas.

Keywords: population dynamics, sardine, stock structure hypothesis, South Africa

Background

The South African sardine stock structure hypothesis has recently been revised (de Moor *et al.* 2023) to consist of two stocks which mix, (i) Cold Temperate Sardine (CTS) which spawn off the west coast and are seldom found off the south coast, but some do (permanently) move via the south coast to take part in the Sardine Run off the east coast and (ii) Warm Temperate Sardine (WTS) which spawn off both the south and (south-)west coasts and which may spend some part of their early life off the west coast before returning to the south coast. This two-area model distinguishes the area west of Cape Agulhas (west coast) from that between Cape Agulhas and Port Alfred (south coast).

There are two key changes in this model from that of de Moor (2024a). First, growth is no longer stock-dependent and is now only coast-dependent, and modelled using a mean length-at-age which accounts for movement from the west to the south coast, instead of keeping track of the age at which WTS moved from the west to the south coast. Secondly, in addition to removing catch by stock off the west coast in proportion to the numbers of exploitable fish available there, the model now also allows for alternative proportions of CTS relative to WTS to be removed for the region to the north of Cape Point and between Cape Point and Cape Agulhas.

Model description

The model is detailed in Appendix A. It has the following key assumptions:

- The two-area two-stock hypothesis models Cold Temperate Sardine (CTS), which spawn off the west coast and Warm Temperate Sardine (WTS), which spawn off both the south and west coasts.
- Annual quarters are 1 Jan – 31 Mar, 1 Apr – 30 Jun, 1 Jul – 30 Sep, 1 Oct – 31 Dec. Quarterly catch is modelled to be taken in a pulse at the mid-point of each quarter.
- Spawning and recruitment are assumed to occur at the start of the 4th quarter (1st October). Growth curves allow for differences in growth by coast and inter-annual variability in this. It is expected that the timing of recruitment will be estimated to be earlier for the south coast, and with faster growth in the first few months on the south coast than the west coast (Sakamoto *et al.* 2020, van der Lingen *et al.* 2023).
- Recruitment of WTS and CTS are modelled independently.

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- All CTS spawning products recruit to the west coast.
- WTS recruit on both the west and south coasts. Some annually-varying proportion of total WTS recruitment is modelled to recruit to the west coast; this proportion reflects a combination of spawning off the west coast as well as spawning products from the south coast that are passively transported to the west coast as recruits.
- Active movement of WTS and CTS from the west to the south coast is assumed to occur at some point after the recruit survey and before the November biomass survey (van der Lingen *et al.* 2023). Active movement is thus modelled to occur at the middle of the 3rd quarter.
- Active movement is modelled to occur before the pulse catch is taken.
- All WTS are assumed to move off the west coast before they turn 3 (van der Lingen *et al.* 2023)¹.
- Growth curves of WTS which recruit to the west coast reflect their growth according to the west coast growth curve for the first part of their life. Growth curves of WTS off the south coast are adjusted at the time of active movement of WTS from the west to the south coast to account for the immigration of WTS sardine which have followed a different growth curve for the first part of their life on the west coast (and are expected to be smaller at the time of movement than WTS of the same cohort on the south coast) (see Appendix B).
- Infection by the *Cardiocephaloides* parasite is assumed to occur in bursts as parasites are released from their subtidal first intermediate host (de Moor *et al.* 2017). This is currently proposed to occur annually at the middle of the 3rd quarter, before active movement. This implies that infected sardine could reach the south coast from ~9 months of age.
- Catch by stock off the south coast is taken in proportion to the exploitable numbers-at-age by stock.
- Directed catch and bycatch off the west coast are calculated separately for the catches taken north of Cape Point (north west) and between Cape Point and Cape Agulhas (south west). These catches can still be modelled to be taken in proportion to the exploitable numbers-at-age by stock off the west coast, with alternatives allowing for a different proportion of CTS : WTS to be taken off the north west and south west coasts. The model is fit to total west coast (and south coast) catch length frequencies, these are not split between the north west and south west regions.
- The directed catch and bycatch modelled to be taken from 1 April to the day before the recruit survey are modelled for two areas only (west coast and south coast), with the catches by stock off the west coast taken in proportion to the exploitable numbers-at-age by stock.
- The November survey estimate of biomass is assumed to correspond with the model estimated biomass at 15th November (mid-point of 4th quarter), as this is the average midpoint of the November surveys (Figure A.1 of de Moor 2022). The 4th quarter catch is modelled to be taken in a pulse immediately before this estimate of biomass is calculated.
- The recruit survey estimate of recruitment is assumed to correspond with the day on which the survey began.
- Both the November and recruit hydroacoustic surveys are assumed to provide a relative index of abundance.

Results from fitting this model to the available data (de Moor *et al.* 2024b) are provided in a separate document (de Moor 2025).

¹ At a later stage, some WTS may need to be modelled to remain on the west coast for a longer period if this assumption does not allow the model to fit the increasing parasite prevalence-at-length rates observed off the south coast.

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Appendix A: Population dynamics equations for a two-stock, two-area model for the South African sardine resource

The assessment is begun at the beginning of the 4th quarter of $y_0 = 1983$, but conditioned to data from $y_1 = 1984$ to $y_n = 2023$. The following subscript/superscript notation is used:

- quarters $q = 1$ denoting January to March, $q = 2$ denoting April to June, $q = 3$ denoting July to September and $q = 4$ denoting October to December;
- ages $a = 0$ to a plus group of $a = 5^+$;
- lengths from a minus group of $l = 2.5^- \text{ cm}$ to a plus group of $l = 24^+ \text{ cm}$;
- stock $j = \text{CTS}$ or $j = \text{WTS}$ denote Cool Temperate Sardine or Warm Temperate Sardine, respectively;
- coast $c = \text{west}$ or $c = \text{south}$ denote the west and south coasts, respectively; and
- infection $p = \text{NI}$ or $p = \text{I}$ denote the sardine uninfected and infected with the digenean ‘tetracotyle-type’ metacercarian endoparasite, respectively.

All parameters are defined in Tables A1 and A2.

Numbers-at-age at 1 January, $y_1 \leq y < y_{n+1}$

$$N_{c,p,y,1,a}^j = \left(N_{c,p,y-1,4,a}^j e^{-M_{c,y,a}^j/8} - C_{c,p,y-1,4,a}^j \right) e^{-M_{c,y,a}^j/8} \quad c = \{w, s\}, p = \{\text{NI}, \text{I}\}, 0 \leq a \leq 5^+ \quad (\text{A1})$$

Numbers-at-age at 1 April ($q = 2$) and 1 July ($q = 3$), $y_1 \leq y < y_n$

$$N_{c,p,y,q,a}^j = \left(N_{c,p,y,q-1,a}^j e^{-M_{c,y,a}^j/8} - C_{c,p,y,q-1,a}^j \right) e^{-M_{c,y,a}^j/8} \quad c = \{w, s\}, p = \{\text{NI}, \text{I}\}, 0 \leq a \leq 5^+ \quad (\text{A2})$$

Recruits at 1 October, $y_0 \leq y < y_n$

$$\begin{aligned} N_{w,\text{NI},y,4,0}^{\text{CTS}} &= R_y^{\text{CTS}} \\ N_{s,\text{NI},y,4,0}^{\text{CTS}} &= 0 \\ N_{w,\text{NI},y,4,0}^{\text{WTS}} &= p_y R_y^{\text{WTS}} \\ N_{s,\text{NI},y,4,0}^{\text{WTS}} &= (1 - p_y) R_y^{\text{WTS}} \\ N_{c,l,y,4,0}^j &= 0 \end{aligned} \quad c = \{w, s\} \quad (\text{A3})$$

Numbers-at-age 1+ at 1 October, $y_1 \leq y < y_n$

$$N_{w,\text{NI},y,4,a}^j = \begin{cases} \left((1 - \text{move}_{y,a-1}^j)(1 - I_y) N_{w,\text{NI},y,3,a-1}^j e^{-M_{w,y,a-1}^j/8} - C_{w,\text{NI},y,3,a-1}^j \right) e^{-M_{w,y,a-1}^j/8} & 1 \leq a \leq 4 \\ \left((1 - \text{move}_{y,4}^j)(1 - I_y) N_{w,\text{NI},y,3,4}^j e^{-M_{w,y,4}^j/8} - C_{w,\text{NI},y,3,4}^j \right) e^{-M_{w,y,4}^j/8} \\ + \left((1 - \text{move}_{y,5+}^j)(1 - I_y) N_{w,\text{NI},y,3,5+}^j e^{-M_{w,y,5+}^j/8} - C_{w,\text{NI},y,3,5+}^j \right) e^{-M_{w,y,5+}^j/8} & a = 5^+ \end{cases}$$

$$N_{w,\text{I},y,4,a}^j = \begin{cases} \left((1 - \text{move}_{y,a-1}^j)(I_y N_{w,\text{NI},y,3,a-1}^j + N_{w,\text{I},y,3,a-1}^j) e^{-M_{w,y,a-1}^j/8} - C_{w,\text{I},y,3,a-1}^j \right) e^{-M_{w,y,a-1}^j/8} & 1 \leq a \leq 4 \\ \left((1 - \text{move}_{y,4}^j)(I_y N_{w,\text{NI},y,3,4}^j + N_{w,\text{I},y,3,4}^j) e^{-M_{w,y,4}^j/8} - C_{w,\text{I},y,3,4}^j \right) e^{-M_{w,y,4}^j/8} \\ + \left((1 - \text{move}_{y,5+}^j)(I_y N_{w,\text{NI},y,3,5+}^j + N_{w,\text{I},y,3,5+}^j) e^{-M_{w,y,5+}^j/8} - C_{w,\text{I},y,3,5+}^j \right) e^{-M_{w,y,5+}^j/8} & a = 5^+ \end{cases}$$

$$N_{s,p,y,4,a}^{\text{CTS}} = 0 \quad p = \{\text{NI}, \text{I}\}, 1 \leq a \leq 5^+ \quad ^2$$

$$N_{s,\text{NI},y,4,a}^{\text{WTS}} = \left(\text{move}_{y,a-1}^{\text{WTS}} (1 - I_y) N_{w,\text{NI},y,3,a-1}^{\text{WTS}} e^{-M_{w,y,a-1}^{\text{WTS}}/8} + N_{s,\text{NI},y,3,a-1}^{\text{WTS}} e^{-M_{s,y,a-1}^{\text{WTS}}/8} - C_{s,\text{NI},y,3,a-1}^{\text{WTS}} \right) e^{-M_{s,y,a-1}^{\text{WTS}}/8} \quad 1 \leq a \leq 4$$

² Uninfected and infected 0-year-old CTS available to the fishery off the south coast at the mid-point of the 3rd quarter will be $\text{move}_{y,0}^j (1 - I_y) N_{w,\text{NI},y,3,0}^{-M_{w,y,0}^{\text{CTS}}/8}$ and $\text{move}_{y,0}^j I_y N_{w,\text{NI},y,3,0}^{-M_{w,y,0}^{\text{CTS}}/8}$, respectively (equation A7).

$$N_{s,l,y,4,a}^{WTS} = \left(move_{y,a-1}^{WTS} (I_y N_{w,NI,y,3,a-1}^{WTS} + N_{w,l,y,3,a-1}^{WTS}) e^{-M_{w,y,a-1}^{WTS}/8} + N_{s,l,y,3,a-1}^{WTS} e^{-M_{s,y,a-1}^{WTS}/8} - C_{s,l,y,3,a-1}^{WTS} \right) e^{-M_{s,y,a-1}^{WTS}/8} \quad 1 \leq a \leq 4$$

$$N_{s,p,y,4,5+}^{WTS} = \left(N_{s,p,y,3,4}^{WTS} e^{-M_{s,y,4}^{WTS}/8} - C_{s,p,y,3,4}^{WTS} \right) e^{-M_{s,y,4}^{WTS}/8} + \left(N_{s,p,y,3,5+}^{WTS} e^{-M_{s,y,5+}^{WTS}/8} - C_{s,p,y,3,5+}^{WTS} \right) e^{-M_{s,y,5+}^{WTS}/8} \quad p = \{NI, I\} \quad (A4)$$

Numbers at length at 1 October, $y_0 \leq y \leq y_n$

The model estimated numbers-at-length range from a 2.5cm minus group to a 24cm plus group, denoted 2.5⁻ and 24⁺, respectively. The model predicted numbers-at-length of ages 1+ only are given by:

$$NL_{c,p,y,l}^{j,1+} = \sum_{a=1}^{5+} A_{c,y,a,l}^{Oct} N_{c,p,y,4,a}^j \quad c = \{w, s\}, p = \{NI, I\}, 2.5^- \text{ cm} \leq l \leq 24^+ \text{ cm} \quad (A5)$$

Spawning biomass at 1 October, $y_0 \leq y \leq y_n$

$$SSB_{c,y}^j = \sum_p \sum_{l=2.5^-}^{24^+} f_{y,l} NL_{c,p,y,l}^{j,1+} w_l \quad c = \{w, s\} \quad (A6)$$

where

$$f_{y,l} = 1 / \left(1 + e^{-(l-L_{50,y})/\delta_y^{mat}} \right) \quad y_0 \leq y \leq y_n \quad (A7)^4$$

Numbers at age mid-way through each quarter, before catch, $y_0 \leq y \leq y_n$

$$N_{mid,c,p,y,q,a}^j = \begin{cases} N_{c,p,y,q,a}^j e^{-M_{c,y,a}^j/8} & q \in \{1,2\} \\ N_{c,p,y,q,a}^j e^{-M_{c,y+1,a}^j/8} & q = 4 \end{cases} \quad c = \{w, s\}, p = \{NI, I\}, 0 \leq a \leq 5^+$$

$$N_{mid,w,NI,y,3,a}^j = (1 - move_{y,a}^j)(1 - I_y) N_{w,NI,y,3,a}^j e^{-M_{w,y,a}^j/8} \quad 0 \leq a \leq 5^+$$

$$N_{mid,w,I,y,3,a}^j = (1 - move_{y,a}^j)(I_y N_{w,NI,y,3,a}^j + N_{w,I,y,3,a}^j) e^{-M_{w,y,a}^j/8} \quad 0 \leq a \leq 5^+$$

$$N_{mid,s,NI,y,3,a}^j = move_{y,a}^j (1 - I_y) N_{w,NI,y,3,a}^j e^{-M_{w,y,a}^j/8} + N_{s,NI,y,3,a}^j e^{-M_{s,y,a}^j/8} \quad 0 \leq a \leq 5^+$$

$$N_{mid,s,I,y,3,a}^j = move_{y,a}^j (I_y N_{w,NI,y,3,a}^j + N_{w,I,y,3,a}^j) e^{-M_{w,y,a}^j/8} + N_{s,I,y,3,a}^j e^{-M_{s,y,a}^j/8} \quad 0 \leq a \leq 5^+ \quad (A8)$$

Numbers at length at 15th November, after catch

$$NL_{mid,c,p,y,l}^j = \begin{cases} \sum_{a=0}^{5+} A_{c,y,a,4,l} (N_{mid,c,p,y,4,a}^j - C_{c,p,y,4,a}^j) & y_0 \leq y \leq y_{n-1} \\ \sum_{a=0}^{5+} A_{c,y+1,a,1,l} \left((N_{mid,c,p,y,4,a}^j - C_{c,p,y,4,a}^j) e^{-M_{w,y+1,a}^j/4} - C_{c,p,y+1,1,a}^j \right) e^{-M_{w,y+1,a}^j/24} & y = y_n \end{cases}$$

$$c = \{w, s\}, p = \{NI, I\}, 2.5^- \text{ cm} \leq l \leq 24^+ \text{ cm} \quad (A9)^5$$

Growth

The proportion of sardine of age a and stock j off coast c that fall in length group l is calculated under the assumption that length-at-age is normally distributed about a von Bertalanffy growth curve. This is calculated at times during the year modelled to correspond to spawning (1 October) and commercial catches (midpoint of each quarter). The equations allow for a change in growth and hence mean length-at-age off the south coast once sardine move from the west coast to the south coast (see Appendix B).

$$A_{c,y,a,l}^{Oct} \sim N(\bar{L}_{c,y,a}^{Oct}, \vartheta_{c,0}^2) \quad c = \{w, s\}, y_0 \leq y \leq y_n, 0 \leq a \leq 5^+, 2.5^- \text{ cm} \leq l \leq 24^+ \text{ cm}^{6,7}$$

where

$$\bar{L}_{w,y,a}^{Oct} = L_{w,\infty} (1 - e^{-\kappa_{w,y-a}(a-t_{0,w})}) \quad 0 \leq a \leq 5^+8$$

$$\bar{L}_{s,y,a}^{Oct} = \begin{cases} L_{s,\infty} (1 - e^{-\kappa_{s,y-a}(a-t_{0,s})}) & a = 0 \\ \bar{L}_{s,y,3,a-1} + (L_{\infty} - \bar{L}_{s,y,3,a-1})(1 - e^{-\kappa_{s,y-a}/8}) & 1 \leq a \leq 5^+ \end{cases} \quad (A10)$$

³ No movement of 3⁺-year olds is modelled.

⁴ Where length is used in an equation, the mid-point of the length class is used.

⁵ This adjustment for $y = y_n$ is to 1st March 2024 for the delayed November 2023 survey which took place from mid-February to mid-March 2024.

⁶ There may be some proportion of this distribution below a length of zero (recruitment after 1st October). In these cases, this proportion is removed from the proportion-at-length of the minus length class.

⁷ The proportion is calculated as the area under the curve between the lower limit and upper limit of length class l . The lower and upper tails are included in the proportions calculated for the minus and plus groups, respectively.

⁸ If the $y - a$ or $y - 1 - a$ subscript of κ is <1983, then 1983 is used.

and $\vartheta_{c,a}$ is given in Appendix B.

$A_{c,y,a,q,l} \sim N(\bar{L}_{c,y,q,a}^{mid}, \vartheta_{c,a,q}^2)$ $c = \{w, s\}, y_0 \leq y \leq y_{n+1}, 1 \leq q \leq 4, 0 \leq a \leq 5^+, 2.5^- \text{ cm} \leq l \leq 24^+ \text{ cm}$ ⁹
where

$$\bar{L}_{w,y,q,a}^{mid} = \begin{cases} L_{w,\infty}(1 - e^{-\kappa_{w,y-a}(a+(2q+1)/8-t_{0,w}})) & 1 \leq q \leq 3 \\ L_{w,\infty}(1 - e^{-\kappa_{w,y-a}(a+1/8-t_{0,w}})) & q = 4 \end{cases}$$

$$\bar{L}_{s,y,q=4,a}^{mid} = \begin{cases} L_{s,\infty}(1 - e^{-\kappa_{s,y-a}(a+1/8-t_{0,s}})) & a = 0 \\ \bar{L}_{s,y,a}^{Oct} + (\bar{L}_{s,y,4,a}^{mid} - \bar{L}_{s,y,a}^{Oct}) = \bar{L}_{s,y,a}^{Oct} + (L_{s,\infty} - \bar{L}_{s,y,a}^{Oct})(1 - e^{-\kappa_{s,y-a}/8}) & 1 \leq a \leq 5^+ \end{cases}$$

$$\bar{L}_{s,y,q=1,a}^{mid} = \begin{cases} L_{s,\infty}(1 - e^{-\kappa_{s,y-a}(a+(2q+1)/8-t_{0,s}})) & a = 0 \\ \bar{L}_{s,y-1,4,a}^{mid} + (\bar{L}_{s,y,1,a}^{mid} - \bar{L}_{s,y-1,4,a}^{mid}) = \bar{L}_{s,y-1,4,a}^{mid} + (L_{s,\infty} - \bar{L}_{s,y-1,4,a}^{mid})(1 - e^{-\kappa_{s,y-a}/4}) & 1 \leq a \leq 5^+ \end{cases}$$

$$\bar{L}_{s,y,q=2,a}^{mid} = \begin{cases} L_{s,\infty}(1 - e^{-\kappa_{s,y-a}(a+(2q+1)/8-t_{0,s}})) & a = 0 \\ \bar{L}_{s,y,1,a}^{mid} + (\bar{L}_{s,y,2,a}^{mid} - \bar{L}_{s,y,1,a}^{mid}) = \bar{L}_{s,y,1,a}^{mid} + (L_{s,\infty} - \bar{L}_{s,y,1,a}^{mid})(1 - e^{-\kappa_{s,y-a}/4}) & 1 \leq a \leq 5^+ \end{cases}$$

$$\bar{L}_{s,y,q=3,a}^{mid} = \frac{\sum_j \sum_p \bar{L}_{w,y,3,a}^{mid} \text{move}_{j,y,a}^j N_{w,p,y,3,a}^j e^{-M_{w,y,a}^j/8} + \bar{L}_{s,y,3,a}^j N_{s,p,y,3,a}^j e^{-M_{s,y,a}^j/8}}{\text{move}_{j,y,a}^j N_{w,p,y,3,a}^j e^{-M_{w,y,a}^j/8} + N_{s,p,y,3,a}^j e^{-M_{s,y,a}^j/8}} \quad 0 \leq a \leq 5^+$$

where

$$\bar{L}_{s,y,q=3,a}^{mid} = \begin{cases} L_{s,\infty}(1 - e^{-\kappa_{s,y-a}(a+(2q+1)/8-t_{0,s}})) & a = 0 \\ \bar{L}_{s,y,2,a}^{mid} + (\bar{L}_{s,y,3,a}^{mid} - \bar{L}_{s,y,2,a}^{mid}) = \bar{L}_{s,y,2,a}^{mid} + (L_{s,\infty} - \bar{L}_{s,y,2,a}^{mid})(1 - e^{-\kappa_{s,y-a}/4}) & 1 \leq a \leq 5^+ \end{cases} \quad (\text{A11})$$

$$\text{with } \kappa_{c,y} = K_c + \varepsilon_{c,y}^\kappa \quad (\text{A12})^{10}$$

$$\text{and } \varepsilon_{c,y}^\kappa = \begin{cases} \eta_{c,y}^\kappa & y = y_0 \\ \rho^\kappa \varepsilon_{c,y-1}^\kappa + \sqrt{1 - (\rho^\kappa)^2} \eta_{c,y}^\kappa & y_1 < y \leq y_n \end{cases} \quad (\text{A13})$$

and $\vartheta_{c,a,q}$ is given in Appendix B.

Biomass associated with the November survey, $y_1 \leq y \leq y_n$

$$B_{c,y} = k_{ac} \sum_j \sum_p \sum_{l=2.5^-}^{24^+} NLmid_{c,p,y,l}^j w_l \quad c = \{w, s\} \quad (\text{A14})$$

Survey trawl selectivity

$$S_l^{\text{survey}} = \begin{cases} 0 & l = 2.5^- \text{ cm} \\ [1 + \exp\{-(l - S_{50})/\delta\}]^{-1} & 3 \text{ cm} \leq l \leq 24^+ \text{ cm} \end{cases} \quad (\text{A15})$$

Proportion-at-length associated with the November survey, $y_1 \leq y \leq y_n$

$$p_{c,y,l} = \begin{cases} \frac{\sum_{l \leq 6 \text{ cm}} \sum_j \sum_p NLmid_{c,p,y,l}^j S_l^{\text{survey}}}{\sum_{l=2.5^-}^{24^+} \sum_j \sum_p NLmid_{c,p,y,l}^j S_l^{\text{survey}}} & l = 6^- \text{ cm} \\ \frac{\sum_j \sum_p NLmid_{c,p,y,l}^j S_l^{\text{survey}}}{\sum_{l=2.5^-}^{24^+} \sum_j \sum_p NLmid_{c,p,y,l}^j S_l^{\text{survey}}} & 6.5 \text{ cm} \leq l \leq 20.5 \text{ cm} \\ \frac{\sum_{l=21}^{23.5} \sum_j \sum_p NLmid_{c,p,y,l}^j S_l^{\text{survey}}}{\sum_{l=2.5^-}^{24^+} \sum_j \sum_p NLmid_{c,p,y,l}^j S_l^{\text{survey}}} & l = 21 - 23.5 \text{ cm} \\ \frac{\sum_j \sum_p NLmid_{c,p,y,24^+}^j S_{24^+}^{\text{survey}}}{\sum_{l=2.5^-}^{24^+} \sum_j \sum_p NLmid_{c,p,y,l}^j S_l^{\text{survey}}} & l = 24^+ \text{ cm} \end{cases} \quad c = \{w, s\} \quad (\text{A16})^{11}$$

Proportion-at-length of fish infected with the parasite in November, $2010 \leq y \leq y_n$

$$P_{c,y,l} = \frac{\sum_j NLmid_{c,l,y,l}^j}{\sum_j \sum_p NLmid_{c,p,y,l}^j} \quad c = \{w, s\}, 7.5 \text{ cm} \leq l \leq 23 \text{ cm} \quad (\text{A17})$$

⁹ The proportion is calculated as the area under the curve between the lower limit and upper limit of length class l . The lower and upper tails are included in the proportions calculated for the minus and plus groups, respectively.

¹⁰ Additive error allows for faster or slower annual growth.

¹¹ The inclusion of model predicted proportion-at-length 24^+ cm is deliberate to take into account the zero samples of 24^+ cm sardine in the survey.

Number of recruits associated with the recruit survey, $1985 \leq y \leq y_n$

$$N_{c,y}^{rec} = k_{c,r} \left(\sum_j N_{c,NI,y,2,0}^j e^{-0.5(1+t_y)M_{c,y,0}^j/12} - C_{c,y}^{bs} \right) e^{-0.5(1+t_y)M_{c,y,0}^j/12} \quad (A18)$$

where

$$M_{c,y,0}^* = \frac{\sum_j M_{c,y,0}^j N_{c,NI,y,2,0}^j e^{-0.5(1+t_y)M_{c,y,0}^j/12}}{\sum_j N_{c,NI,y,2,0}^j e^{-0.5(1+t_y)M_{c,y,0}^j/12}} \quad (A19a)$$

$$k_{w,r} = k_{cov} \times k_{ac} \quad (A19a)$$

$$k_{s,r} = k_{covS} \times k_{cov} \times k_{ac} \quad (A19b)$$

Commercial selectivity, $y_0 \leq y \leq y_{n+1}$

$$S_{c,y,q,l} = \begin{cases} 0 & l \leq 5.5cm \\ \chi_{c,y,q} \exp \left\{ -\frac{(l-\bar{l}_{1,y})^2}{(\sigma_1^{sel})^2} \right\} + \frac{1}{1 + \exp \{ -(l-\bar{l}_{2,c,y,q})/(\sigma_2^{sel})^2 \}} & 6cm \leq l \leq l_{max} = 23cm^{12} \\ S_{c,y,q,lmax} & l > l_{max} \end{cases} \quad c = \{w, s\}, 1 \leq q \leq 4 \quad (A20a)$$

with $S_{c,1983,4,l} = S_{c,1984,4,l}$, and

$$S_{c,y,q,a} = \sum_{l=2.5}^{24^+} A_{c,y,a,q,l} S_{c,y,q,l} \quad c = \{w, s\}, 1 \leq q \leq 4, 0 \leq a \leq 5^+ \quad (A20b)$$

Bycatch in the anchovy directed fishery, $y_1 \leq y \leq y_{n+1}$

$$C_{c,p,y,q,a}^{by,j} = Nmid_{c,p,y,q,a}^j F_{c,y,q,a}^{j,By} \quad c = \{w, s\}, p = \{NI, I\}, 1 \leq q \leq 4, 0 \leq a \leq 1 \quad (A21)^{13}$$

Catch in the directed sardine and round herring bycatch fisheries, $y_0 \leq y \leq y_{n+1}$

$$C_{c,p,y,q,a}^{dir,j} = (Nmid_{c,p,y,q,a}^j - C_{c,p,y,q,a}^{by,j}) S_{c,y,q,a} F_{c,y,q}^j \quad c = \{w, s\}, p = \{NI, I\}, 1 \leq q \leq 4, 0 \leq a \leq 5^+ \quad (A22)$$

Total catch, $y_0 \leq y \leq y_{n+1}$

$$C_{c,p,y,q,a}^j = C_{c,p,y,q,a}^{by,j} + C_{c,p,y,q,a}^{dir,j} \quad c = \{w, s\}, p = \{NI, I\}, 1 \leq q \leq 4, 0 \leq a \leq 5^+ \quad (A23)$$

Fished proportion of the available age 0 and 1 biomass from the sardine bycatch with the anchovy directed fishery, $y_1 \leq y \leq y_{n+1}$

$$\begin{aligned} F_{s,y,q=1,a=0}^{j,By} &= F_{s,y,q=1,a=0}^{By} = \frac{\sum_{m=1}^3 \sum_{l < lcut_{y,m}} ByC_{s,y,m,l}^{RLF}}{\sum_j \sum_p Nmid_{c=p,y,q=1,a=0}^j} & F_{s,y,1,1}^{j,By} &= F_{s,y,q=1,a=1}^{By} = \frac{\sum_{m=1}^3 \sum_{l \geq lcut_{y,m}} ByC_{s,y,m,l}^{RLF}}{\sum_j \sum_p Nmid_{c=p,y,q=1,a=1}^j} \\ F_{s,y,2,0}^{j,By} &= F_{s,y,2,0}^{By} = \frac{\sum_{m=4}^6 \sum_{l < lcut_{y,m}} ByC_{s,y,m,l}^{RLF}}{\sum_j \sum_p Nmid_{c=p,y,2,0}^j} & F_{s,y,2,1}^{j,By} &= F_{s,y,2,1}^{By} = \frac{\sum_{m=4}^6 \sum_{l \geq lcut_{y,m}} ByC_{s,y,m,l}^{RLF}}{\sum_j \sum_p Nmid_{c=p,y,2,1}^j} \\ F_{s,y,3,0}^{j,By} &= F_{s,y,3,0}^{By} = \frac{\sum_{m=7}^9 \sum_{l < lcut_{y,m}} ByC_{s,y,m,l}^{RLF}}{\sum_j \sum_p Nmid_{c=p,y,3,0}^j} & F_{s,y,3,1}^{j,By} &= F_{s,y,3,1}^{By} = \frac{\sum_{m=7}^9 \sum_{l \geq lcut_{y,m}} ByC_{s,y,m,l}^{RLF}}{\sum_j \sum_p Nmid_{c=p,y,3,1}^j} \\ F_{s,y,4,0}^{j,By} &= F_{s,y,4,0}^{By} = \frac{\sum_{m=10}^{12} \sum_{l < lcut_{y,m}} ByC_{s,y,m,l}^{RLF}}{\sum_j \sum_p Nmid_{c=p,y,4,0}^j} & F_{s,y,4,1}^{j,By} &= F_{s,y,4,1}^{By} = \frac{\sum_{m=10}^{12} \sum_{l \geq lcut_{y,m}} ByC_{s,y,m,l}^{RLF}}{\sum_j \sum_p Nmid_{c=p,y,4,1}^j} \end{aligned} \quad (A24a)$$

$$\begin{aligned} F_{w,y,1,0}^{CTS,By} &= \sum_{m=1}^3 \sum_{l < lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_1 Nmid_{w,p,y,1,0}^{WTS} + Nmid_{w,p,y,1,0}^{CTS}} \right] + \sum_{m=1}^3 \sum_{l < lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\sum_p \Phi_2 Nmid_{w,p,y,1,0}^{CTS} + Nmid_{w,p,y,1,0}^{WTS}} \right] \\ F_{w,y,1,1}^{CTS,By} &= \sum_{m=1}^3 \sum_{l \geq lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_1 Nmid_{w,p,y,1,1}^{WTS} + Nmid_{w,p,y,1,1}^{CTS}} \right] + \sum_{m=1}^3 \sum_{l \geq lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\sum_p \Phi_2 Nmid_{w,p,y,1,1}^{CTS} + Nmid_{w,p,y,1,1}^{WTS}} \right] \\ F_{w,y,2,0}^{CTS,By} &= \sum_{m=4}^6 \sum_{l < lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_1 Nmid_{w,p,y,2,0}^{WTS} + Nmid_{w,p,y,2,0}^{CTS}} \right] + \sum_{m=4}^6 \sum_{l < lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\sum_p \Phi_2 Nmid_{w,p,y,2,0}^{CTS} + Nmid_{w,p,y,2,0}^{WTS}} \right] \\ F_{w,y,2,1}^{CTS,By} &= \sum_{m=4}^6 \sum_{l \geq lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_1 Nmid_{w,p,y,2,1}^{WTS} + Nmid_{w,p,y,2,1}^{CTS}} \right] + \sum_{m=4}^6 \sum_{l \geq lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\sum_p \Phi_2 Nmid_{w,p,y,2,1}^{CTS} + Nmid_{w,p,y,2,1}^{WTS}} \right] \\ F_{w,y,3,0}^{CTS,By} &= \sum_{m=7}^9 \sum_{l < lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_1 Nmid_{w,p,y,3,0}^{WTS} + Nmid_{w,p,y,3,0}^{CTS}} \right] + \sum_{m=7}^9 \sum_{l < lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\sum_p \Phi_2 Nmid_{w,p,y,3,0}^{CTS} + Nmid_{w,p,y,3,0}^{WTS}} \right] \\ F_{w,y,3,1}^{CTS,By} &= \sum_{m=7}^9 \sum_{l \geq lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_1 Nmid_{w,p,y,3,1}^{WTS} + Nmid_{w,p,y,3,1}^{CTS}} \right] + \sum_{m=7}^9 \sum_{l \geq lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\sum_p \Phi_2 Nmid_{w,p,y,3,1}^{CTS} + Nmid_{w,p,y,3,1}^{WTS}} \right] \end{aligned}$$

¹² Normalised to a maximum of 1.

¹³ "Selectivity" is incorporated in $F_{c,y,q,a}^{By}$, as the sardine bycatch is typically independent of sardine abundance, but rather correlated with anchovy recruitment which varies from year to year.

$$\begin{aligned}
F_{w,y,4,0}^{CTS,By} &= \sum_{m=10}^{12} \sum_{l < lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_1 N_{mid_{w,p,y,4,0}}^{WTS} + N_{mid_{w,p,y,4,0}}^{CTS}} \right] + \sum_{m=10}^{12} \sum_{l < lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\sum_p \Phi_2 N_{mid_{w,p,y,4,0}}^{CTS} + N_{mid_{w,p,y,4,0}}^{WTS}} \right] \\
F_{w,y,4,1}^{CTS,By} &= \sum_{m=10}^{12} \sum_{l \geq lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_1 N_{mid_{w,p,y,4,1}}^{WTS} + N_{mid_{w,p,y,4,1}}^{CTS}} \right] + \sum_{m=10}^{12} \sum_{l \geq lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\sum_p \Phi_2 N_{mid_{w,p,y,4,1}}^{CTS} + N_{mid_{w,p,y,4,1}}^{WTS}} \right] \\
F_{w,y,1,0}^{WTS,By} &= \sum_{m=1}^3 \sum_{l < lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\sum_p \Phi_1 N_{mid_{w,p,y,1,0}}^{WTS} + N_{mid_{w,p,y,1,0}}^{CTS}} \right] + \sum_{m=1}^3 \sum_{l < lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_2 N_{mid_{w,p,y,1,0}}^{CTS} + N_{mid_{w,p,y,1,0}}^{WTS}} \right] \\
F_{w,y,1,1}^{WTS,By} &= \sum_{m=1}^3 \sum_{l \geq lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\sum_p \Phi_1 N_{mid_{w,p,y,1,1}}^{WTS} + N_{mid_{w,p,y,1,1}}^{CTS}} \right] + \sum_{m=1}^3 \sum_{l \geq lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_2 N_{mid_{w,p,y,1,1}}^{CTS} + N_{mid_{w,p,y,1,1}}^{WTS}} \right] \\
F_{w,y,2,0}^{WTS,By} &= \sum_{m=4}^6 \sum_{l < lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\sum_p \Phi_1 N_{mid_{w,p,y,2,0}}^{WTS} + N_{mid_{w,p,y,2,0}}^{CTS}} \right] + \sum_{m=4}^6 \sum_{l < lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_2 N_{mid_{w,p,y,2,0}}^{CTS} + N_{mid_{w,p,y,2,0}}^{WTS}} \right] \\
F_{w,y,2,1}^{WTS,By} &= \sum_{m=4}^6 \sum_{l \geq lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\sum_p \Phi_1 N_{mid_{w,p,y,2,1}}^{WTS} + N_{mid_{w,p,y,2,1}}^{CTS}} \right] + \sum_{m=4}^6 \sum_{l \geq lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_2 N_{mid_{w,p,y,2,1}}^{CTS} + N_{mid_{w,p,y,2,1}}^{WTS}} \right] \\
F_{w,y,3,0}^{WTS,By} &= \sum_{m=7}^9 \sum_{l < lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\sum_p \Phi_1 N_{mid_{w,p,y,3,0}}^{WTS} + N_{mid_{w,p,y,3,0}}^{CTS}} \right] + \sum_{m=7}^9 \sum_{l < lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_2 N_{mid_{w,p,y,3,0}}^{CTS} + N_{mid_{w,p,y,3,0}}^{WTS}} \right] \\
F_{w,y,3,1}^{WTS,By} &= \sum_{m=7}^9 \sum_{l \geq lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\sum_p \Phi_1 N_{mid_{w,p,y,3,1}}^{WTS} + N_{mid_{w,p,y,3,1}}^{CTS}} \right] + \sum_{m=7}^9 \sum_{l \geq lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_2 N_{mid_{w,p,y,3,1}}^{CTS} + N_{mid_{w,p,y,3,1}}^{WTS}} \right] \\
F_{w,y,4,0}^{WTS,By} &= \sum_{m=10}^{12} \sum_{l < lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\sum_p \Phi_1 N_{mid_{w,p,y,4,0}}^{WTS} + N_{mid_{w,p,y,4,0}}^{CTS}} \right] + \sum_{m=10}^{12} \sum_{l < lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_2 N_{mid_{w,p,y,4,0}}^{CTS} + N_{mid_{w,p,y,4,0}}^{WTS}} \right] \\
F_{w,y,4,1}^{WTS,By} &= \sum_{m=10}^{12} \sum_{l \geq lcut_{y,m}} ByC_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\sum_p \Phi_1 N_{mid_{w,p,y,4,1}}^{WTS} + N_{mid_{w,p,y,4,1}}^{CTS}} \right] + \sum_{m=10}^{12} \sum_{l \geq lcut_{y,m}} ByC_{NW,y,m,l}^{RLF} \left[\frac{1}{\sum_p \Phi_2 N_{mid_{w,p,y,4,1}}^{CTS} + N_{mid_{w,p,y,4,1}}^{WTS}} \right]
\end{aligned} \tag{A24b}$$

A penalty is imposed within the model to ensure that $F_{c,y,q,a}^{j,By} < 0.95$.

Fished proportion of the available biomass from the directed sardine catch and sardine bycatch with round herring fishery,
 $y_0 \leq y \leq y_{n+1}$

Defining exploitable numbers-at-age by stock and coast as: $N_{c,p,y,q}^{j,exp} = \sum_p \sum_{a=0}^{5+} (N_{mid_{c,p,y,q,a}}^j - C_{c,p,y,q,a}^{by,j}) S_{c,y,q,a}$

$$\begin{aligned}
F_{s,y,q=1}^j &= F_{s,y,q=1} = \frac{\sum_{m=1}^3 \sum_{l \geq 6cm} C_{s,y,m,l}^{RLF}}{\sum_j N_{c=s,y,q=1}^{j,exp}} \\
F_{s,y,2}^j &= F_{s,y,2} = \frac{\sum_{m=4}^6 \sum_{l \geq 6cm} C_{s,y,m,l}^{RLF}}{\sum_j N_{s,y,2}^{j,exp}} \\
F_{s,y,3}^j &= F_{s,y,3} = \frac{\sum_{m=7}^9 \sum_{l \geq 6cm} C_{s,y,m,l}^{RLF}}{\sum_j N_{s,y,3}^{j,exp}} \\
F_{s,y,4}^j &= F_{s,y,4} = \frac{\sum_{m=10}^{12} \sum_{l \geq 6cm} C_{s,y,m,l}^{RLF}}{\sum_j N_{s,y,4}^{j,exp}} \\
F_{w,y,q=1}^{CTS} &= \sum_{m=1}^3 \sum_{l \geq 6cm} C_{SW,y,m,l}^{RLF} \left[\frac{1}{\Phi_1 N_{w,y,1}^{WTS,exp} + N_{w,y,1}^{CTS,exp}} \right] + \sum_{m=1}^3 \sum_{l \geq 6cm} C_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\Phi_2 N_{w,y,1}^{CTS,exp} + N_{w,y,1}^{WTS,exp}} \right] \\
F_{w,y,2}^{CTS} &= \sum_{m=4}^6 \sum_{l \geq 6cm} C_{SW,y,m,l}^{RLF} \left[\frac{1}{\Phi_1 N_{w,y,2}^{WTS,exp} + N_{w,y,2}^{CTS,exp}} \right] + \sum_{m=4}^6 \sum_{l \geq 6cm} C_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\Phi_2 N_{w,y,2}^{CTS,exp} + N_{w,y,2}^{WTS,exp}} \right] \\
F_{w,y,3}^{CTS} &= \sum_{m=7}^9 \sum_{l \geq 6cm} C_{SW,y,m,l}^{RLF} \left[\frac{1}{\Phi_1 N_{w,y,3}^{WTS,exp} + N_{w,y,3}^{CTS,exp}} \right] + \sum_{m=7}^9 \sum_{l \geq 6cm} C_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\Phi_2 N_{w,y,3}^{CTS,exp} + N_{w,y,3}^{WTS,exp}} \right] \\
F_{w,y,4}^{CTS} &= \sum_{m=10}^{12} \sum_{l \geq 6cm} C_{SW,y,m,l}^{RLF} \left[\frac{1}{\Phi_1 N_{w,y,4}^{WTS,exp} + N_{w,y,4}^{CTS,exp}} \right] + \sum_{m=10}^{12} \sum_{l \geq 6cm} C_{NW,y,m,l}^{RLF} \left[\frac{\Phi_2}{\Phi_2 N_{w,y,4}^{CTS,exp} + N_{w,y,4}^{WTS,exp}} \right] \tag{A25a} \\
F_{w,y,q=1}^{WTS} &= \sum_{m=1}^3 \sum_{l \geq 6cm} C_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\Phi_1 N_{w,y,1}^{WTS,exp} + N_{w,y,1}^{CTS,exp}} \right] + \sum_{m=1}^3 \sum_{l \geq 6cm} C_{NW,y,m,l}^{RLF} \left[\frac{1}{\Phi_2 N_{w,y,1}^{CTS,exp} + N_{w,y,1}^{WTS,exp}} \right] \\
F_{w,y,2}^{WTS} &= \sum_{m=4}^6 \sum_{l \geq 6cm} C_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\Phi_1 N_{w,y,2}^{WTS,exp} + N_{w,y,2}^{CTS,exp}} \right] + \sum_{m=4}^6 \sum_{l \geq 6cm} C_{NW,y,m,l}^{RLF} \left[\frac{1}{\Phi_2 N_{w,y,2}^{CTS,exp} + N_{w,y,2}^{WTS,exp}} \right] \\
F_{w,y,3}^{WTS} &= \sum_{m=7}^9 \sum_{l \geq 6cm} C_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\Phi_1 N_{w,y,3}^{WTS,exp} + N_{w,y,3}^{CTS,exp}} \right] + \sum_{m=7}^9 \sum_{l \geq 6cm} C_{NW,y,m,l}^{RLF} \left[\frac{1}{\Phi_2 N_{w,y,3}^{CTS,exp} + N_{w,y,3}^{WTS,exp}} \right] \\
F_{w,y,4}^{WTS} &= \sum_{m=10}^{12} \sum_{l \geq 6cm} C_{SW,y,m,l}^{RLF} \left[\frac{\Phi_1}{\Phi_1 N_{w,y,4}^{WTS,exp} + N_{w,y,4}^{CTS,exp}} \right] + \sum_{m=10}^{12} \sum_{l \geq 6cm} C_{NW,y,m,l}^{RLF} \left[\frac{1}{\Phi_2 N_{w,y,4}^{CTS,exp} + N_{w,y,4}^{WTS,exp}} \right] \tag{A25b}
\end{aligned}$$

A penalty is imposed within the model to ensure that $\max_a \{S_{c,y,q,a}^j\} F_{c,y,q}^j < 0.95$. Fish <6cm were seldom¹⁴ caught and were thus not used in fitting this model. Commercial selectivity-at-length is fixed to zero for length classes <6cm (equation A20a).

Catch-at-length from the directed and round herring bycatch fisheries, $y_1 \leq y \leq y_{n+1}$

¹⁴ Less than 3% of the quarters north of Cape Point, less than 1% of the quarters between Cape Point and Cape Agulhas or east of Cape Agulhas, and less than 3% of the quarters for the whole coast.

$$C_{c,y,q,l}^{dir} = \sum_j \sum_p \sum_{a=0}^{5+} (Nmid_{c,p,y,q,a}^j - C_{c,p,y,q,a}^{by,j}) A_{c,y,a,q,l} S_{c,y,q,l} F_{c,y,q}^j$$

$$c = \{w, s\}, 1 \leq q \leq 3, 6cm \leq l \leq 24^+cm \text{ (A26)}^{15}$$

Proportion-at-length associated with the directed catch and round herring bycatch, $y_1 \leq y \leq y_n$

$$p_{c,y,q,l}^{coml} = \begin{cases} \frac{C_{c,y,q,l}^{dir}}{\sum_{l=6}^{24^+} C_{c,y,q,l}^{dir}} & 6cm \leq l \leq 22.5cm \\ \frac{\sum_{l=23}^{24^+} C_{c,y,q,l}^{dir}}{\sum_{l=6}^{24^+} C_{c,y,q,l}^{dir}} & l = 23^+cm \end{cases} \quad c = \{w, s\}, 1 \leq q \leq 4 \text{ (A27)}^{11}$$

Fitting the Model to Observed Data (Likelihood)

$$-\ln L = -\ln L^{Nov} - \ln L^{rec} - \ln L^{sur\ prop} - \ln L^{com\ prop} - \ln L^{prev} \quad (\text{A28})$$

where

$$-\ln L^{Nov} = 0.5 \sum_c \sum_{y=y_1}^{y_n} \left\{ \frac{5^5 \left(\frac{|\ln(\hat{B}_{c,y}) - \ln(B_{c,y})|}{\sqrt{\sigma_{c,y,Nov}^2 + \phi_{ac}^2 + \lambda_{c,N}^2}} \right)^5}{5^5 + \left(\frac{|\ln(\hat{B}_{c,y}) - \ln(B_{c,y})|}{\sqrt{\sigma_{c,y,Nov}^2 + \phi_{ac}^2 + \lambda_{c,N}^2}} \right)^5} \right\}^{2/5} + \ln[2\pi(\sigma_{c,y,Nov}^2 + \phi_{ac}^2 + \lambda_{c,N}^2)] \quad (\text{A29})$$

$$-\ln L^{rec} = 0.5 \sum_c \sum_{y=y_2}^{y_n} \left\{ \frac{5^5 \left(\frac{|\ln(\hat{N}_{c,y}^{rec}) - \ln(N_{c,y}^{rec})|}{\sqrt{\sigma_{c,y,rec}^2 + \phi_{ac}^2 + \lambda_{c,r}^2}} \right)^5}{5^5 + \left(\frac{|\ln(\hat{N}_{c,y}^{rec}) - \ln(N_{c,y}^{rec})|}{\sqrt{\sigma_{c,y,rec}^2 + \phi_{ac}^2 + \lambda_{c,r}^2}} \right)^5} \right\}^{2/5} + \ln[2\pi(\sigma_{c,y,rec}^2 + \phi_{ac}^2 + \lambda_{c,r}^2)] \quad (\text{A30})$$

$$-\ln L^{sur\ prop} = w_{prop}^{sur} \sum_c \sum_{y=y_1}^{y_n} \left\{ \sum_{l=6}^{21^+} \left\{ \frac{\left(\sqrt{\hat{p}_{c,y,l}} - \sqrt{p_{c,y,l}} \right)^2}{2\sigma_{c,sur}^2} + \ln(\sigma_{c,sur}) \right\} + \frac{\left(0 - \sqrt{p_{c,y,24^+}} \right)^2}{2\sigma_{c,sur}^2} + \ln(\sigma_{c,sur}) \right\} \quad (\text{A31})^{16}$$

$$-\ln L^{com\ prop} = 0.5 w_{prop}^{com} \sum_j \sum_{y=y_1}^{1986} \sum_{q=1}^4 \sum_{l=6}^{23^+} \left\{ \frac{\left(\sqrt{\hat{p}_{c,y,q,l}^{coml}} - \sqrt{p_{c,y,q,l}^{coml}} \right)^2}{2\sigma_{c,com}^2} + \ln(\sigma_{c,com}) \right\} +$$

$$w_{prop}^{com} \sum_j \sum_{y=1987}^{y_n} \sum_{q=1}^4 \sum_{l=6}^{23^+} \left\{ \frac{\left(\sqrt{\hat{p}_{c,y,q,l}^{coml}} - \sqrt{p_{c,y,q,l}^{coml}} \right)^2}{2\sigma_{c,com}^2} + \ln(\sigma_{c,com}) \right\} \quad (\text{A32})$$

$$-\ln L^{prev} = \sum_c \sum_{y=2010}^{y_n} \sum_{l=7.5cm}^{23cm} -n_{c,y,l}^{prev} \ln(P_{c,y,l}) - (N_{c,y,l}^{prev} - n_{c,y,l}^{prev}) \ln(1 - P_{c,y,l}) \quad (\text{A33})$$

A “robustified likelihood” is used for the contributions from the hydro-acoustic surveys to ensure no undue influence from any extreme (outlying) values for residuals. The functional form chosen to “robustify” makes negligible difference for standardised residuals of magnitude three or less, but essentially treats large standardised residuals as if they do not exceed five in magnitude.

¹⁵ Note the model predicted commercial catch of lengths <6cm is zero, from a zero commercial selectivity in equation A20. This is consistent with the range of length classes in the observed commercial proportions-at-lengths.

¹⁶ The 21⁺ group in this equation consists of the length classes 21cm, 21.5cm, 22cm, 22.5cm, 23cm and 23.5cm.

Table A1. Assessment model parameters and variables with associated fixed values or prior distributions and, for derived variables, associated equation numbers. As the majority of

Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation
Annual numbers and biomass	$N_{c,p,y,q,a}^j$	Billions	$N_{s,NI,1983,4,1}^{CTS} = 0$ $N_{w,NI,1983,4,1}^{CTS}, N_{s,NI,1983,4,1}^{WTS}$ $N_{w,NI,1983,4,1}^{WTS} \sim U(0,50)$ $N_{c,NI,1983,4,a}^j = 0, 2 \leq a \leq 5^+$ $N_{c,I,1983,4,a}^j = 0, 0 \leq a \leq 5^+$ $R_y^{CTS} / 10 \sim U(0,50), R_{c,y}^{WTS} / 10 \sim U(0,50)$	A1 – A4
	R_y^j			
	p_y	-	$\sim U(0,1)$	
	$M_{c,y,a}^j$	Year ⁻¹	$M_{w,y,0}^{CTS} = 1.0^{17}, M_{w,y,1^+}^{CTS} = 1.0$ $M_{s,y,0}^{WTS} = M_{w,y,0}^{CTS} + M_{juv}^{diff}$ $M_{juv}^{diff} \sim U(-0.5,0.5)$ $M_{s,y,1^+}^{WTS} = M_{s,y,0}^{WTS} - M_{ad}^{diff}, M_{ad}^{diff} \sim U(0,0.5)$ $M_{s,y,a}^{CTS} = M_{w,y,a}^{CTS} + M_{inc}^{CTS}, M_{w,y,a}^{WTS} = M_{s,y,a}^{WTS} + M_{inc}^{WTS}, 0 \leq a \leq 5^+, M_{inc}^j \sim U(0,1)$ $M_{c,1983,a}^j = M_{c,1984,a}^j, M_{c,2024,a}^j = M_{c,2023,a}^j$ $move_{y,0}^{CTS} = move_0^{CTS} \sim 1 / (1 + e^{(move_0^{CTS} - 0.1) / 0.1})$ $move_{y,1^+}^{CTS} = 0$ $move_{y,0}^{WTS} \sim Beta(1.05,1.05)$ $move_{y,1}^{WTS} = \phi \times move_{y,0}^{WTS}, \phi \sim U(0,1)$ $move_{y,2}^{WTS} = 1$ $move_{y,3^+}^{WTS} = NA$	
	$move_{y,a}^j$	-		
	I_y	-	$= 0, y_1 \leq y \leq 2007$ $\sim U(0,1), 2008 \leq y \leq y_n$	
	$N_{mid,c,p,y,q,a}^j$	Billions		A8
	w_l	Grams	$0.0071118 \times l^{3.181}^{18}$	

¹⁷ The 1.0 year⁻¹ values were selected based on previous assessments.¹⁸ A coast-wide length-weight relationship was calculated from samples collected from November surveys and taken to apply both to the time of spawning and November survey (OLSPS 2023). This relationship is currently assumed to additionally apply to commercial catches. A robustness test will investigate the impact of using different relationships for the commercial catches for the west and south coasts; the latter estimated from commercial samples from the south coast is substantially different from the November survey relationship.

Table A1 (Continued).

Parameter / Variable		Description	Units / Scale	Fixed Value / Prior Distribution	Equation
Annual #s and biomass	$f_{y,l}$	Proportion of sardine of length l (in cm) that are mature in year y	-		A7
	$L_{50,y}$	Length at 50% maturity in year y	Cm	Table A.3	
	δ_y^{mat}	Rate of increase in maturity at length in year y	-	Table A.3	
	$SSB_{c,y}^j$	Spawning biomass of stock j on coast c at the beginning of October in year y	Thousand tons		A6
	$B_{c,y}$	Total biomass on coast c at the middle of November in year y , associated with the November survey	Thousand tons		A14
	$N_{c,y}^{rec}$	Number of recruits on coast c at the time of the recruit survey in year y	Billions		A18
Growth and proportions-at-length	$NL_{c,p,y,l}^{j,1+}$	Numbers-at-length l of ages 1+ of stock j on coast c at 1 October in year y that are uninfected ($p = NI$) or infected ($p = I$) with the endoparasite	Billions		A5
	$NL_{c,p,y,l}^{mid,j}$	Numbers-at-length l of stock j on coast c on 15 th November in year y that are uninfected ($p = NI$) or infected ($p = I$) with the endoparasite	Billions		A9
	$\bar{L}_{c,y,a}^{Oct}$	Mean length of age a sardine on coast c at 1 October in year y	Cm		A10
	$\bar{L}_{c,y,q,a}^{mid}$	Mean length of age a sardine on coast c mid-way through quarter q of year y	Cm		A11
	$\hat{L}_{s,y,q=3,a}$	Mean length of age a sardine on the south coast before movement mid-way through quarter 3 of year y	Cm		A11
	$A_{c,y,a,l}^{Oct}$	Proportion of age a sardine on coast c that falls in the length group l at 1 October in year y	-		A10
	$A_{c,y,a,q,l}$	Proportion of age a sardine on coast c that falls in the length group l mid-way through quarter q of year y	-		A11
	$\kappa_{c,y}$	Somatic growth rate parameter for sardine on coast c of recruit cohort in year y	Year ⁻¹		A12
	K_c	Average somatic growth rate parameter for sardine on coast c	Year ⁻¹	$\kappa_w \sim U(0,2), \kappa_s - \kappa_w \sim U(0,2)$	
	$L_{c,\infty}$	Maximum length (in expectation) of sardine on coast c	Cm	$L_{c,\infty} = \frac{L_{c,a=1}}{1 - e^{-\kappa_c(1-t_{0,c})}}$ where $L_{w,1} \sim U(5,18), L_{s,1} - L_{w,1} \sim U(0,13)$	

Table A1 (Continued).

Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation
Growth and proportions-at-length	$t_{0,c}$	Year	$t_{0,s}, t_{0,w} \sim U(-0.5, 0.5)$	A13
	$\varepsilon_{c,y}^k$			
	$\eta_{c,y}^k$		$N(0, 0.2^2)$	
	ρ^k		$U(-1, 1)$	
	ϑ_a	-	$\vartheta_a/3 \sim \text{Beta}(1.05, 1.05), a = 0, 1, 2^{+19}$	A16
	ϑ_{bf}		$\vartheta_{bf} \sim \text{Beta}(1.05, 1.05)$	
	ϑ_{af}		$\vartheta_{af} \sim \text{Beta}(1.05, 1.05)$	
	$p_{c,y,l}$	-		
	$P_{c,y,l}$			A17
	$p_{c,y,l}^{coml}$	-		A27
Selectivity	S_l^{survey}	-		A15 ²⁰
	S_{50}	Cm	$\sim 1/(1 + e^{(S_{50}-7)/0.5})$	
	δ	-	$U(0.005, 5)$	
	$S_{c,y,q,l}$	-		A20a ²¹
	$S_{c,y,q,a}$	-	$S_{c,1983,4,a} = S_{c,1984,4,a}$	
	$\chi_{c,y,q}$	-	$\chi_{w,y,q} \sim U(0, 1), \chi_{s,y,q} = 0^{22}$, with $\chi_{w,y,1} = \chi_{w,1}$ and $\chi_{w,y,4} = \chi_{w,q}$	A20b
	$\bar{l}_{1,y}$	cm	$N(10, 1^2)$	
	$\bar{l}_{2,c,y,q}$	cm	$\bar{l}_{2,c,y,1} - \bar{l}_{1,y} \sim U(0, 15)^{23}$ $\bar{l}_{2,c,y,3} - \bar{l}_{1,y} \sim U(0, 15)$	
	$(\sigma_1^{sel})^2$	cm	$U(2, 15)$	
	$(\sigma_2^{sel})^2$	cm	$U(0, 10)$	

¹⁹ Upper bound of 3 precludes unrealistically large lengths for young fish.²⁰ Some smaller fish escape through the trawl net.²¹ $S_{c,y,1,l} = S_{c,y-1,4,l}$ and $S_{c,y,2,l} = S_{c,y,3,l}$.²² No bycatch is modelled for the South Coast.²³ Estimated for two time periods per stock: 1984-1993, 1994-2023 (west) and 1984-1997, 1998-2023 (south).

Table A1 (Continued).

Parameter / Variable		Description	Units / Scale	Fixed Value / Prior Distribution	Equation
Survey bias	k_{ac}	Multiplicative bias associated with the hydro-acoustic survey	-	$\ln(k_{ac}^S) \sim N(-0.224, 0.211^2)$ ²⁴	A19
	$k_{c,r}$	Multiplicative bias associated with the recruit survey on coast c	-		
	k_{cov}	Multiplicative bias associated with the coverage of the recruits during the recruit survey in comparison to the coverage of the biomass during the November survey	-	$U(0,1)$	
	k_{covS}	Multiplicative bias associated with the coverage of the recruits on the South Coast in comparison to the West Coast during the recruit survey		$= 0.3 + 0.7 \times \text{Beta}(1.05, 1.05)$ ²⁵	
Catch	$C_{c,p,y,q,a}^j$	Total number of age a fish of stock j caught on coast c during quarter q of year y that are uninfected ($p = NI$) or infected ($p = I$) with the endoparasite	Billions		A23
	$C_{c,p,y,q,a}^{by,j}$	Number of age a fish of stock j bycaught in the anchovy-directed fishery on coast c in quarter q of year y that are uninfected ($p = NI$) or infected ($p = I$) with the endoparasite	Billions		A21
	$C_{c,p,y,q,a}^{dir,j}$	Number of age a fish of stock j caught in the sardine-directed and round herring bycatch fisheries on coast c in quarter q of year y that are uninfected ($p = NI$) or infected ($p = I$) with the endoparasite	Billions		A22
	$C_{c,y,q,l}^{dir}$	Number of length l fish of stock j caught in the sardine-directed and round herring bycatch fisheries on coast c in quarter q of year y	Billions		A26
	$F_{c,y,q,a}^{By}$	Fished proportion of bycatch in the anchovy-directed fishery in quarter q of year y for age class a on coast c	-		A24
	$F_{c,y,q}$	Fished proportion of the directed and round herring bycatch fisheries in quarter q of year y for a fully selected age class a on coast c	-		A25
	ϕ_1	Relative weighting towards WTS in catches off the south west coast	-	Different fixed values	
	ϕ_2	Relative weighting towards CTS in catches off the north west coast	-	for alternative models	

²⁴ de Moor *et al.* (2024a).

²⁵ Lower bound of 0.3 selected in discussions with scientists on these surveys and their field experience.

Table A1 (Continued).

Parameter / Variable	Description	Units / Scale	Fixed Value / Prior Distribution	Equation
$-\ln L^{Nov}$	Contribution to the negative log likelihood from the model fit to the November survey biomass data	-		A29
$-\ln L^{rec}$	Contribution to the negative log likelihood from the model fit to the recruit survey data	-		A30
$-\ln L^{surpropl}$	Contribution to the negative log likelihood from the model fit to the November survey proportion-at-length data	-		A31
$-\ln L^{compropl}$	Contribution to the negative log likelihood from the model fit to the quarterly commercial proportion-at-length data	-		A32
$-\ln L^{prev}$	Contribution to the negative log likelihood from the model fit to the November parasite prevalence-at-length data	-		A33
ϕ_{ac}	CV associated with factors which cause bias in the acoustic survey estimates and which vary inter-annually rather than remain fixed over time	-	=0.198 ²⁶	
$(\lambda_{c,N/r})^2$	Additional variance (over and above $\sigma_{c,y,Nov/rec}^2$ and ϕ_{ac}^2) associated with the November/recruit surveys on coast c	-	0 ²⁷	
w_{propl}^{sur}	Weighting applied to the remaining survey proportion-at-length data	-	$= 0.5 \times 0.167$ ²⁸	
$\sigma_{c,sur}^2$	Variance associated with the survey proportion-at-length data on coast c	-	$\frac{\sum_{y=y_1}^{y_n} \sum_{l=6}^{21^+} \left(\sqrt{\hat{p}_{c,y,l}} - \sqrt{p_{c,y,l}} \right)^2}{\sum_{y=y_1}^{y_n} \sum_{l=6}^{21^+} 1}$ ²⁹	
w_{propl}^{com}	Weighting applied to the commercial proportion-at-length data	-	$= 0.5 \times 0.04$ ³⁰	
$\sigma_{c,com}^2$	Variance associated with the commercial proportion-at-length data on coast c	-	$\frac{0.5 \times \sum_{y=y_1}^{1986} \sum_{q=1}^4 \sum_{l=6}^{23^+} \left(\sqrt{\hat{p}_{c,y,q,l}^{coml}} - \sqrt{p_{c,y,q,l}^{coml}} \right)^2 + \sum_{y=1987}^{y_n} \sum_{q=1}^4 \sum_{l=6}^{23^+} \left(\sqrt{\hat{p}_{c,y,q,l}^{coml}} - \sqrt{p_{c,y,q,l}^{coml}} \right)^2}{0.5 \times \sum_{y=y_1}^{1986} \sum_{q=1}^4 \sum_{l=6}^{23^+} 1 + \sum_{y=1987}^{y_n} \sum_{q=1}^4 \sum_{l=6}^{23^+} 1}$ ³¹	

²⁶ de Moor *et al.* (2024a).²⁷ $U(0,5)$ will be estimated in robustness tests.²⁸ To allow for autocorrelation, based upon data being available ~6 times more frequently than annual age data which contain maximum information content on this.²⁹ The 21⁺ group in this equation consists of the length classes 21cm, 21.5cm, 22cm, 22.5cm, 23cm and 23.5cm.³⁰ To allow for autocorrelation, based upon data being available ~4x6 times more frequently than annual age data which contain maximum information content on this.³¹ A shorter range of lengths is used for the south coast given the near absence of data outside this range, resulting in small/zero residuals, which would negatively bias this estimate.

Table A2. Assessment model data, detailed in de Moor *et al.* (2024b).

Quantity	Description	Units / Scale
t_y	Time lapsed between 1 May and the start of the recruit survey in year y	Months
$C_{c,y}^{bs}$	Number of juveniles caught between 1 April and the day before the start of the recruit survey on coast c in year y	Billions
$C_{c,y,m,l}^{RLF}$	Number of fish in length class l landed by the directed sardine fishery or as bycatch with round herring (1984-2011) or ≥ 14 cm sardine bycatch (2012-2023) in month m of year y on coast c	Billions
$ByC_{c,y,m,l}^{RLF}$	Number of fish in length class l landed as bycatch with anchovy (1984-2011) or ≤ 14 cm sardine bycatch (2012-2023) in month m of year y on coast c	Billions
$lcut_{y,m}$	Cut off length for recruits in the bycatch with anchovy in month m of year y	Cm
$\hat{B}_{c,y}$	Acoustic survey estimate of biomass from the November survey on coast c in year y	Thousand tons
$\sigma_{c,y,Nov}$	Survey sampling CV associated with $\hat{B}_{c,y}$ that reflects survey inter-transect variance	-
$\hat{N}_{c,y}^{rec}$	Acoustic survey estimate of recruitment from the recruit survey on coast c in year y	Billions
$\sigma_{c,y,rec}$	Survey sampling CV associated with $\hat{N}_{c,y}^{rec}$ that reflects survey inter-transect variance	-
$\hat{p}_{c,y,l}$	Observed proportion (by number) in length group l in the November survey on coast c of year y	-
$\hat{p}_{c,y,q,l}^{coml}$	Observed proportion (by number) of the directed catch and round herring bycatch of fish in length group l on coast c during quarter q of year y	-
$n_{c,y,l}^{prev}$	Number of sardine in length class l sampled from the November survey on coast c in year y that were tested and found to be infected with the endoparasite	Numbers
$N_{c,y,l}^{prev}$	Number of sardine in length class l sampled from the November survey on coast c in year y that were tested for infection with the endoparasite	Numbers

Table A.3. Annual length (in cm) at which maturity is 50% and associated steepness (Yemane *et al.* 2024).

Year	$L_{50,y}$	δ_y^{mat}	Year	$L_{50,y}$	δ_y^{mat}	Year	$L_{50,y}$	δ_y^{mat}
1984 ³²	13.451	1.287	1998	15.581	2.857	2012	12.577	3.426
1985	13.451	1.287	1999	15.447	2.043	2013	11.224	3.938
1986	11.305	0.563	2000	15.829	2.907	2014	12.228	2.313
1987	9.828	1.597	2001	14.914	1.967	2015	11.178	2.297
1988	9.872	1.789	2002	15.710	2.256	2016	12.125	1.990
1989	12.042	1.642	2003	14.898	2.743	2017	12.259	1.943
1990	11.848	1.582	2004	14.002	2.245	2018	11.358	1.729
1991	9.569	2.379	2005	13.370	2.002	2019	11.117	1.402
1992	14.037	1.440	2006	14.234	1.205	2020	11.910	2.049
1993	12.996	1.982	2007	13.571	1.629	2021 ³³	12.095	1.753
1994	13.040	1.606	2008	11.498	2.649	2022	12.280	1.456
1995	15.480	2.237	2009	13.429	2.117	2023 ³⁴	11.785	1.716
1996	14.953	2.087	2010	13.651	2.644			
1997	15.318	2.808	2011	12.542	4.404			

³² No data available. Set equal to 1985.³³ No data available. Used average 2020, 2022.³⁴ No data for November 2023 available. Used average 2017-2020, 2022.

Appendix B: Mean length at age and associated variance for fish which move from the west to the south coastBackground

The ‘underlying base’ lengths-at-age ($\tilde{L}_{c,y,a}$) for the west and south coasts (and independent of stock) at the time of year when the sardine age (1 October) are given by a von Bertalanffy growth curve:

$$\tilde{L}_{c,y,a} = L_{c,\infty}(1 - e^{-\kappa_{c,y-a}(a-t_{0,c})}) \quad c \in \{w,s\}, 0 \leq a \leq 5^+ \quad (B1)$$

and in the middle of each quarter:

$$\tilde{L}_{c,y,q,a} = \begin{cases} L_{c,\infty}(1 - e^{-\kappa_{c,y-1-a}(a+(2q+1)/8-t_{0,c})}) & 1 \leq q \leq 3 \\ L_{c,\infty}(1 - e^{-\kappa_{c,y-a}(a+1/8-t_{0,c})}) & q = 4 \end{cases} \quad 0 \leq a \leq 5^+ \quad (B2)$$

The growth, or change in length, between two time periods for sardine growing according to a particular von Bertalanffy growth curve can be written as (ignoring coast or quarter dependencies for now):

$$L_{a+\Delta} - L_a = L_{\infty}(1 - e^{-\kappa(a+\Delta-t_0)}) - L_{\infty}(1 - e^{-\kappa(a-t_0)}) = L_{\infty}e^{-\kappa(a-t_0)}(1 - e^{-\Delta\kappa}) = (L_{\infty} - L_a)(1 - e^{-\Delta\kappa}) \quad (B3)$$

The sardine which move from the west to the south coast do so in the middle of the 3rd quarter before age 3. The length-at-age on the south coast is adjusted to take account of the sardine which move from the west coast, which may be of a different length for the same age³⁵. This adjustment uses the mean length-at-age weighted by the number of sardine-at-age originally found off the south coast and those moving to the west coast at age a . The mean length-at-age at the middle of the 3rd quarter is thus given by:

$$\frac{\sum_j \sum_p \tilde{L}_{w,y,a} \text{move}_{y,a}^j N_{w,p,y,3,a}^j e^{-M_{w,y,a}^j/8} + \tilde{L}_{s,y,a} N_{s,p,y,3,a}^j e^{-M_{s,y,a}^j/8}}{\text{move}_{y,a}^j N_{w,p,y,3,a}^j e^{-M_{w,y,a}^j/8} + N_{s,p,y,3,a}^j e^{-M_{s,y,a}^j/8}} \quad (B4)$$

While the equation above allows for both WTS and CTS, in practice the mean length is calculated using WTS only for the following reasons (so that the summation over stock j is restricted to WTS only in equation (B9)). There are no CTS already present off the south coast at the start of the 3rd quarter; only those which move from the west coast would be included in the equation above. The CTS which move from the west to the south coast are a small proportion of the WTS that move (van der Lingen *et al.* 2023, Figure 13a of de Moor 2024b), and form a very small proportion of the 3rd quarter catch off the south coast (Figure 22 of de Moor 2024b). Furthermore, CTS are modelled to be distributed only off the south coast in the middle of the 3rd quarter before they are assumed to move further eastwards, outside of the modelled area (van der Lingen *et al.* 2023, Teske *et al.* 2021), thus leaving only the WTS that were originally off the south coast at the start of the 3rd quarter and the WTS which moved from the west coast at the middle of the 3rd quarter.

The variance around the ‘base underlying’ length-at-age at 1 October is coast- and stock-independent and given by ϑ_a^2 , and the variance at the middle of each quarter is given by $\left[\left(1 - \frac{(2q+1)}{8}\right)\vartheta_a + \frac{(2q+1)}{8}\vartheta_{a+1}\right]^2$ for $1 \leq q \leq 3$ and $\left[\left(1 - \frac{1}{8}\right)\vartheta_a + \frac{1}{8}\vartheta_{a+1}\right]^2$ for $q = 4$.

Mean length at age for combined distributions

The mean lengths-at-age for sardine off the south coast at 1 October are thus given by:

$$\bar{L}_{s,y,a}^{Oct} = \begin{cases} L_{s,\infty}(1 - e^{-\kappa_{s,y-a}(a-t_{0,s})}) & a = 0 \\ \bar{L}_{s,y,q=3,a-1} + (\bar{L}_{s,y,a} - \bar{L}_{s,y,q=3,a-1}) & 1 \leq a \leq 5^+ \end{cases}$$

which, using equation (B3), can be given as

$$\bar{L}_{s,y,a}^{Oct} = \begin{cases} L_{s,\infty}(1 - e^{-\kappa_{s,y-a}(a-t_{0,s})}) & a = 0 \\ \bar{L}_{s,y,3,a-1} + (L_{s,\infty} - \bar{L}_{s,y,3,a-1})(1 - e^{-\kappa_{s,y-a}/8}) & 1 \leq a \leq 5^+ \end{cases} \quad (B5)$$

The mean lengths-at-age for sardine off the south coast at the middle of each subsequent quarter are then given by:

$$\bar{L}_{s,y,q=4,a}^{mid} = \begin{cases} L_{s,\infty}(1 - e^{-\kappa_{s,y-a}(a+1/8-t_{0,s})}) & a = 0 \\ \bar{L}_{s,y,a}^{Oct} + (\bar{L}_{s,y,4,a}^{mid} - \bar{L}_{s,y,a}^{Oct}) = \bar{L}_{s,y,a}^{Oct} + (L_{s,\infty} - \bar{L}_{s,y,a}^{Oct})(1 - e^{-\kappa_{s,y-a}/8}) & 1 \leq a \leq 5^+ \end{cases} \quad (B6)$$

³⁵ Sardine moving from the west coast are expected to be smaller based on the assumption that fish off the south coast are expected to (a) grow faster in their first few months of life (Sakamoto *et al.* 2020) and (b) be spawned earlier than those off the west coast (van der Lingen *et al.* 2023).

$$\bar{L}_{s,y,q=1,a}^{mid} = \begin{cases} L_{s,\infty} (1 - e^{-\kappa_{s,y-1-a}(a+(2q+1)/8-t_{0,s})}) & a = 0 \\ \bar{L}_{s,y-1,4,a}^{mid} + (\bar{L}_{s,y,1,a}^{mid} - \bar{L}_{s,y-1,4,a}^{mid}) = \bar{L}_{s,y-1,4,a}^{mid} + (L_{s,\infty} - \bar{L}_{s,y-1,4,a}^{mid})(1 - e^{-\kappa_{s,y-1-a}/4}) & 1 \leq a \leq 5^+ \end{cases} \quad (B7)$$

$$\bar{L}_{s,y,q=2,a}^{mid} = \begin{cases} L_{s,\infty} (1 - e^{-\kappa_{s,y-1-a}(a+(2q+1)/8-t_{0,s})}) & a = 0 \\ \bar{L}_{s,y,1,a}^{mid} + (\bar{L}_{s,y,2,a}^{mid} - \bar{L}_{s,y,1,a}^{mid}) = \bar{L}_{s,y,1,a}^{mid} + (L_{s,\infty} - \bar{L}_{s,y,1,a}^{mid})(1 - e^{-\kappa_{s,y-1-a}/4}) & 1 \leq a \leq 5^+ \end{cases} \quad (B8)$$

$$\bar{L}_{s,y,q=3,a}^{mid} = \frac{\sum_p (\bar{L}_{w,y,3,a}^{mid} move_{y,a}^{WTS} N_{w,p,y,3,a}^{WTS} e^{-M_{w,y,a}^{WTS}/8} + \bar{L}_{s,y,3,a}^{mid} N_{s,p,y,3,a}^{WTS} e^{-M_{s,y,a}^{WTS}/8})}{\sum_p (move_{y,a}^{WTS} N_{w,p,y,3,a}^{WTS} e^{-M_{w,y,a}^{WTS}/8} + N_{s,p,y,3,a}^{WTS} e^{-M_{s,y,a}^{WTS}/8})} \quad 0 \leq a \leq 5^+ \quad (B9)$$

where

$$\hat{L}_{s,y,q=3,a} = \begin{cases} L_{s,\infty} (1 - e^{-\kappa_{s,y-1-a}(a+(2q+1)/8-t_{0,s})}) & a = 0 \\ \bar{L}_{s,y,2,a}^{mid} + (\bar{L}_{s,y,3,a}^{mid} - \bar{L}_{s,y,2,a}^{mid}) = \bar{L}_{s,y,2,a}^{mid} + (L_{s,\infty} - \bar{L}_{s,y,2,a}^{mid})(1 - e^{-\kappa_{s,y-1-a}/4}) & 1 \leq a \leq 5^+ \end{cases} \quad (B10)$$

Variance about the combined distribution's mean length at age

While the equations above can be used for the west coast, the variance for the combined length distribution off the south coast needs to be calculated about the 'combined mean length' rather than the original 'base underlying' length-at-age. The variance around the combined mean length at the middle of the 3rd quarter off the south coast is given by³⁷:

$$Var(\bar{L}_{s,y,3,a}^{mid}) = o_w Var(\bar{L}_{w,y,3,a}^{mid}) + o_s Var(\hat{L}_{s,y,3,a}) + o_w (\bar{L}_{w,y,3,a}^{mid})^2 + o_s (\hat{L}_{s,y,3,a})^2 - (o_w \bar{L}_{w,y,3,a}^{mid} + o_s \hat{L}_{s,y,3,a})^2$$

where the weightings for each of the original length-at-age distributions are given by:

$$o_w = \frac{\sum_p move_{y,a}^{WTS} N_{w,p,y,3,a}^{WTS} e^{-M_{w,y,a}^{WTS}/8}}{\sum_p (move_{y,a}^{WTS} N_{w,p,y,3,a}^{WTS} e^{-M_{w,y,a}^{WTS}/8} + N_{s,p,y,3,a}^{WTS} e^{-M_{s,y,a}^{WTS}/8})} \text{ and } o_s = \frac{\sum_p N_{s,p,y,3,a}^{WTS} e^{-M_{s,y,a}^{WTS}/8}}{\sum_p (move_{y,a}^{WTS} N_{w,p,y,3,a}^{WTS} e^{-M_{w,y,a}^{WTS}/8} + N_{s,p,y,3,a}^{WTS} e^{-M_{s,y,a}^{WTS}/8})}.$$

However, given the potentially multi-modal distribution is being approximated by a normal distribution (equations A10, A11) and also given estimation complications with solving for the variance about the combined mean length, the following simplified method was used (Table B1).

- i) On the west coast, variance between 1 October of one year to the next is calculated using linear interpolation (as has previously been assumed, e.g. de Moor (2024a))
- ii) On the south coast, variance at the middle of the 3rd quarter after movement is estimated to be greater than it would have been prior to movement.
- iii) On the south coast, variance between mid-September of one year to the next is calculated using linear interpolation.

³⁶ As mentioned above, for practical purposes, this mean is calculated for WTS only.

³⁷ <https://stats.stackexchange.com/questions/16608/what-is-the-variance-of-the-weighted-mixture-of-two-gaussians>

Table B1. Variance assumed about the mean length at age. The five estimable parameters are given in **bold**. As there is no further movement after age 3 and the variance on the west coast is estimated to be the same for ages 2+, there is no change to the variance from age 3 for the south coast.

Age	Time	West Coast	South Coast
0	1 Oct	$\vartheta_{w,0} = \boldsymbol{\vartheta_0}$	$\vartheta_{s,0} = \boldsymbol{\vartheta_0}$
0	Mid-Q4	$\vartheta_{w,0,4} = \left(1 - \frac{1}{8}\right)\boldsymbol{\vartheta_0} + \frac{1}{8}\boldsymbol{\vartheta_1}$	$\vartheta_{s,0,4} = \left(1 - \frac{1}{8}\right)\boldsymbol{\vartheta_0} + \frac{1}{8}\boldsymbol{\vartheta_1}$
0	Mid-Q1	$\vartheta_{w,0,1} = \left(1 - \frac{(2q+1)}{8}\right)\boldsymbol{\vartheta_0} + \frac{(2q+1)}{8}\boldsymbol{\vartheta_1}$	$\vartheta_{s,0,1} = \left(1 - \frac{(2q+1)}{8}\right)\boldsymbol{\vartheta_0} + \frac{(2q+1)}{8}\boldsymbol{\vartheta_1}$
0	Mid-Q2	$\vartheta_{w,0,2} = \left(1 - \frac{(2q+1)}{8}\right)\boldsymbol{\vartheta_0} + \frac{(2q+1)}{8}\boldsymbol{\vartheta_1}$	$\vartheta_{s,0,2} = \left(1 - \frac{(2q+1)}{8}\right)\boldsymbol{\vartheta_0} + \frac{(2q+1)}{8}\boldsymbol{\vartheta_1}$
0	Mid-Q3	$\vartheta_{w,0,3} = \left(1 - \frac{(2q+1)}{8}\right)\boldsymbol{\vartheta_0} + \frac{(2q+1)}{8}\boldsymbol{\vartheta_1}$	$\vartheta_{s,0,3} = \vartheta_{w,0,3} + \boldsymbol{\vartheta_{bf}} + \boldsymbol{\vartheta_{af}}$
1	1 Oct	$\vartheta_{w,1} = \boldsymbol{\vartheta_1}$	$\vartheta_{s,1} = \frac{1}{8}\boldsymbol{\vartheta_{s,1,3}^{bf\ move}} + \frac{7}{8}\boldsymbol{\vartheta_{s,0,3}}$
1	Mid-Q4	$\vartheta_{w,1,4} = \left(1 - \frac{1}{8}\right)\boldsymbol{\vartheta_1} + \frac{1}{8}\boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,1,4} = \frac{1}{4}\boldsymbol{\vartheta_{s,1,3}^{bf\ move}} + \frac{3}{4}\boldsymbol{\vartheta_{s,0,3}}$
1	Mid-Q1	$\vartheta_{w,1,1} = \left(1 - \frac{(2q+1)}{8}\right)\boldsymbol{\vartheta_1} + \frac{(2q+1)}{8}\boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,1,1} = \frac{2}{4}\boldsymbol{\vartheta_{s,1,3}^{bf\ move}} + \frac{2}{4}\boldsymbol{\vartheta_{s,0,3}}$
1	Mid-Q2	$\vartheta_{w,1,2} = \left(1 - \frac{(2q+1)}{8}\right)\boldsymbol{\vartheta_1} + \frac{(2q+1)}{8}\boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,1,2} = \frac{3}{4}\boldsymbol{\vartheta_{s,1,3}^{bf\ move}} + \frac{1}{4}\boldsymbol{\vartheta_{s,0,3}}$
1	Mid-Q3	$\vartheta_{w,1,3} = \left(1 - \frac{(2q+1)}{8}\right)\boldsymbol{\vartheta_1} + \frac{(2q+1)}{8}\boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,1,3} = \boldsymbol{\vartheta_{s,1,3}^{bf\ move}} + \boldsymbol{\vartheta_{af}}$ $\boldsymbol{\vartheta_{s,1,3}^{bf\ move}} = \vartheta_{w,1,3} + \boldsymbol{\vartheta_{bf}}$
2	1 Oct	$\vartheta_{w,2} = \boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,2} = \frac{1}{8}\boldsymbol{\vartheta_{s,2,3}^{bf\ move}} + \frac{7}{8}\boldsymbol{\vartheta_{s,1,3}}$
2	Mid-Q4	$\vartheta_{w,2,4} = \boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,2,4} = \frac{1}{4}\boldsymbol{\vartheta_{s,2,3}^{bf\ move}} + \frac{3}{4}\boldsymbol{\vartheta_{s,1,3}}$
2	Mid-Q1	$\vartheta_{w,2,1} = \boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,2,1} = \frac{2}{4}\boldsymbol{\vartheta_{s,2,3}^{bf\ move}} + \frac{2}{4}\boldsymbol{\vartheta_{s,1,3}}$
2	Mid-Q2	$\vartheta_{w,2,2} = \boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,2,2} = \frac{3}{4}\boldsymbol{\vartheta_{s,2,3}^{bf\ move}} + \frac{1}{4}\boldsymbol{\vartheta_{s,1,3}}$
2	Mid-Q3	$\vartheta_{w,2,3} = \boldsymbol{\vartheta_{2+}}$	$\vartheta_{s,2,3} = \boldsymbol{\vartheta_{s,2,3}^{bf\ move}} + \boldsymbol{\vartheta_{af}}$ $\boldsymbol{\vartheta_{s,2,3}^{bf\ move}} = \vartheta_{w,2,3} + \boldsymbol{\vartheta_{bf}}$
3+		$\boldsymbol{\vartheta_{2+}}$	$\boldsymbol{\vartheta_{s,2,3}}$